

Parenthood Timing and Gender Inequality

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Motivation

Gender inequality in OECD labor markets typically emerges with parenthood

- Understanding the role of parenthood is crucial for addressing gender inequity

Quantifying the effects of parenthood is challenging due to selection:

- Parents may differ systematically from those who are and/or remain childless

Addressing selection requires quasi-experimental fertility variation:

- A setting where pregnancy is as-good-as-random (various quasi-experiments)
- Compare those who conceive at that moment to those who do not

But many women who do not conceive initially eventually become mothers

- Existing evidence identifies a weighted average of parenthood and timing effects
- Or restricts timing effects to identify parenthood effects

These forces may work in opposite directions and require distinct policies

This Paper

What are the effects of parenthood and its timing?

- How would outcomes differ if individuals remained childless or delayed childbirth?
- 1. New approach to estimate treatment effects in sequential quasi-experiments
 - Settings where previously unassigned individuals can repeatedly undergo assignment
 - E.g. a series of conception attempts
- 2. Empirical evidence using novel Dutch administrative data
 - Focus on couples undergoing intrauterine insemination
 - Common and minimally invasive infertility treatment
- 3. Quantify and compare the effects of parenthood and its timing
 - How does parenthood shape women's outcomes and gender inequality?
 - Does later childbearing mitigate or amplify losses?

Preview of Main Results

- Parenthood persistently reduces women's work hours and income
 - Yearly reductions of 16% and 18%, respectively
- Parenthood causes a large share, but far from all, of post-child gender inequality
 - Around 45% in work hours and 35% in income
- Delayed childbearing results in smaller losses
 - 2.5-year delay lowers four-year post-birth losses by 12% for earnings and 23% for hours

Literature and Contribution

1. Existing quasi-experimental studies focus on parenthood effects

- Hotz et al. (2005); Agüero & Marks (2008); Cristia (2008); Miller (2011); Lundborg et al. (2017); Bensnes et al. (2023); Gallen et al. (2023); Lundborg et al. (2024)

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3. Separating treatment and timing effects in quasi-experiments is a common challenge

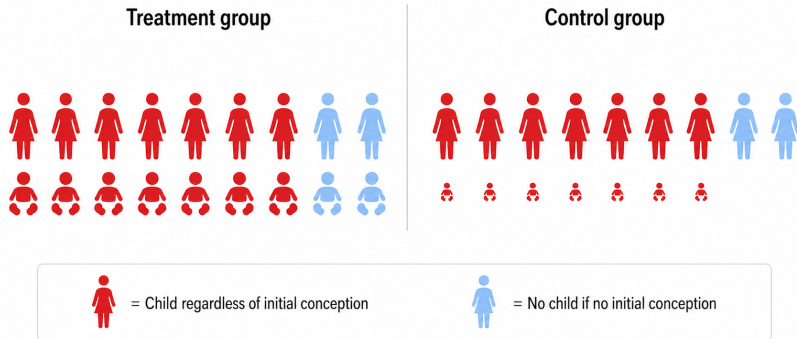
- Education programs with multiple admission cycles, assignment to judges, promotion tournaments

Method applicable to many other settings with sequential quasi-experiments

Typical Quasi-Experiment

At a given moment, some women conceive (treatment), others do not (control).

- Control group: remain childless or conceive later



Mothers differ in age at first birth, career stage at birth, and potentially # of children.

- Comparison mixes motherhood vs. childlessness with earlier vs. later motherhood

Baseline Reduced-Form Setup

Address selection using a quasi-experiment: women undergoing artificial insemination.

- Outcome of the first procedure $Z_1 \in \{0, 1\}$
- Parenthood indicator $D \in \{0, 1\}$
- Potential labor market outcomes:
 - Y^E : mother early (first IUI), Y^L : mother later (after first IUI), Y^N : remain childless
- Effect of parenthood: $Y^E - Y^N$
- Effect of parenthood timing: $Y^L - Y^E$

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Standard reduced form argument:

1. Assume success of the first procedure is random (conditional on age)
2. Compare women whose first procedure succeeds to those whose fails:

$$\mathbb{E}[\text{Parenthood effect} \mid D = 0 \text{ if } Z_1 = 0]c - \mathbb{E}[\text{Timing effect} \mid D = 1 \text{ if } Z_1 = 0](1 - c)$$

- $c = \Pr(\text{no child if } Z_1 = 0) \approx 0.25$ in practice

Method Overview

Key observation: many women conceive through subsequent IUIs.

- Variation in later IUI success could help separate treatment and timing effects
- Leveraging this variation is challenging because the number of IUIs is endogenous

Result 1: Selection into subsequent IUIs can be addressed using the observed IUI #

- Identify average outcomes in the scenario where entire IUI sequence fails
- Many women remain childless; some conceive (much later) via natural means

Result 2: Worst-case bounds allow to separate treatment and timing effects

- Bound the effect of conceiving at the first IUI vs. remaining childless
- Bound the effect of conceiving at the first IUI vs. after the IUI sequence

No assumptions required on selection into subsequent IUIs or on the structure of treatment/timing effects within or across individuals.

Statistical Model

Women differ in two unobserved characteristics:

- “Willingness” to undergo IUIs, $W \in \{1, \dots, \overline{w}\}$
 - Would undergo W IUIs for the first child if all IUIs failed
- “Reliance” on IUIs, $R \in \{0, 1\}$
 - No child if all IUIs fail, $R = 1$

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Observables:

- IUI j success indicator, Z_j , for procedures before having any children
- Number of realized IUIs:

$$A = \min(\{j : Z_j = 1\} \cup \{W\})$$

- Parenthood indicator:

$$D = \max(Z_A, 1 - R)$$

Sequential Unconfoundedness

Assumption (Sequential Unconfoundedness)

$$(Y^E, Y^N, R, W) \perp\!\!\!\perp Z_j | A \geq j.$$

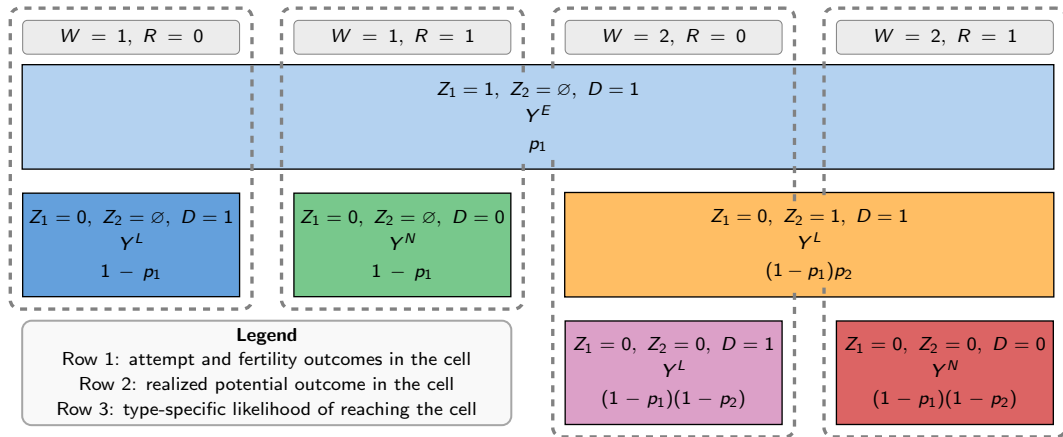
In words: once sperm at IUI j are implanted, whether this results in a conception is as-good-as-random

- **The decision to undergo the procedure can be endogenous**
- Y^E, Y^N, R and W can be related
- Main method relaxes to covariate-conditional version: age at procedure, technology

Alternatively: R and W may correlate with Z_j 's, but not with potential outcomes.

- E.g. sterility may affect R and W , but should not affect potential outcomes

Identification Intuition: Procedure Failure Scenario

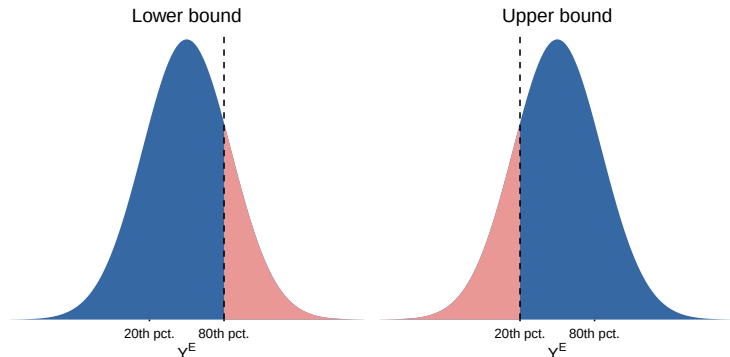


Identify:

- Relier childless outcomes: $\mathbb{E}[Y^L | W = w, R = 0]$
- Non-relier later motherhood outcomes: $\mathbb{E}[Y^L | W = w, R = 0]$
- Type shares: $\Pr(W = w, R = r)$

Identification Intuition: First Procedure Success Scenario

1. Treated group is a representative sample but their types are unobserved
2. Identify $\Pr(R = 1) = 0.8$ on control group
3. Assume most extreme distributions of types in treated group
4. Bound $\mathbb{E}[Y^E | R = 1]$

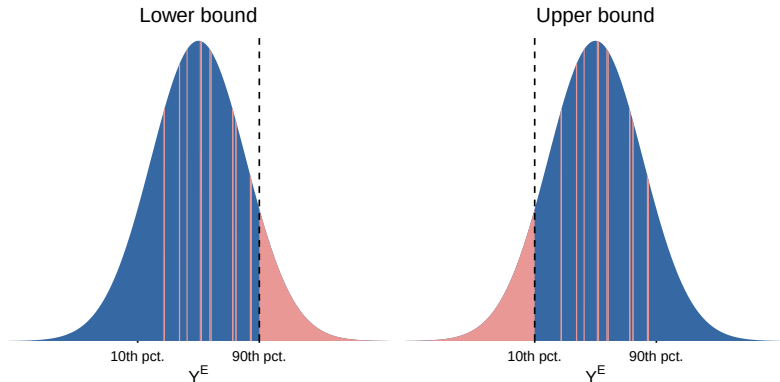


Argument for timing effects/non-reliers is symmetric.

Narrowing the Bounds Further

Use additional information which on mother are reliers:

- Some women have non-IUI children after IUI succeeds
- May be reasonable to assume they are not reliant on IUIs
- Consistent with being determined to have at least one child
- Reduces uncertainty around which women are reliers



Point Identification

Remaining uncertainty:

- Among women with no natural conception after IUI success, which would have remained childless had IUI failed?

In the IUI sample, non-IUI conception may largely be a matter of chance:

- If so, women with no natural conception after IUI success can be compared directly with women who remain childless after the IUI sequence fails

Testable implication:

- Pre-IUI observables should be balanced between these two groups

Further Technical Details

Formal identification →

- Covariate-conditional sequential unconfoundedness: age and procedure type
- Combine the steps in a semi-parametric moment equation

Using covariates to narrow the bounds →

- The bounds are sharp

Inference complicated by trimming of the outcome distribution →

- Build on a double/debiased machine learning approach by Semenova (2023)
- Construct orthogonal moment functions that are robust to first-stage nonparametric estimation errors in quantile and other nuisance functions

Background and Data

Assisted conception procedures →

- In-vitro fertilization: invasive medical procedure, first 3 free
- Intrauterine insemination: direct sperm injection, minimally invasive, free

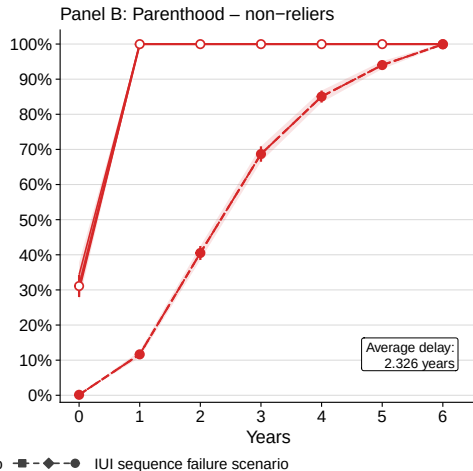
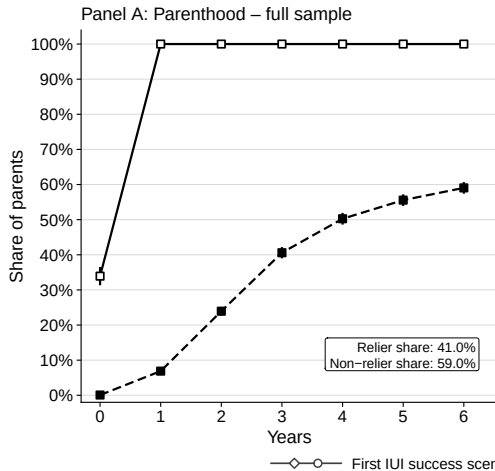
Dutch family policies and labor market similar to OECD average →

- 16 weeks maternity + pregnancy leave, 1 week paternity leave

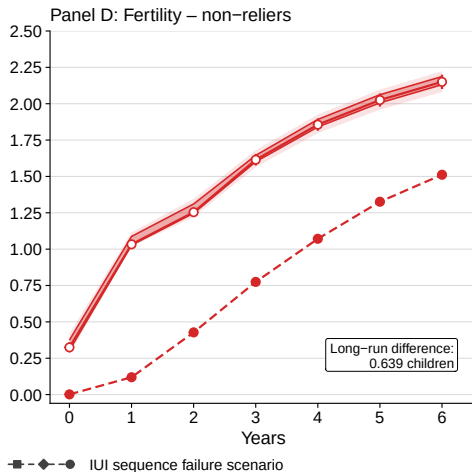
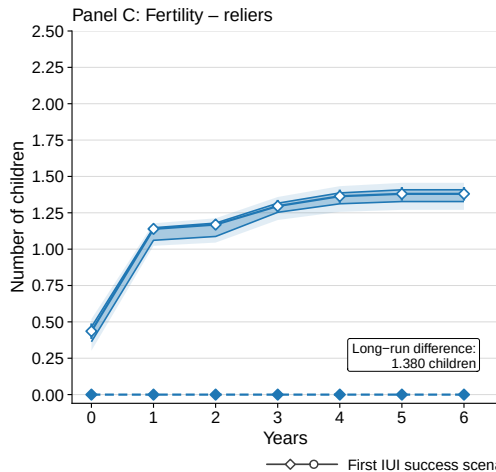
Data combining ACP medical records with tax records →

- Work hours and income include leave; results for hours corrected for uncertainty
- 12,892 cohabiting opposite-sex couples
- Balance: ACP success at each attempt uncorr. with past outcomes cond. on age

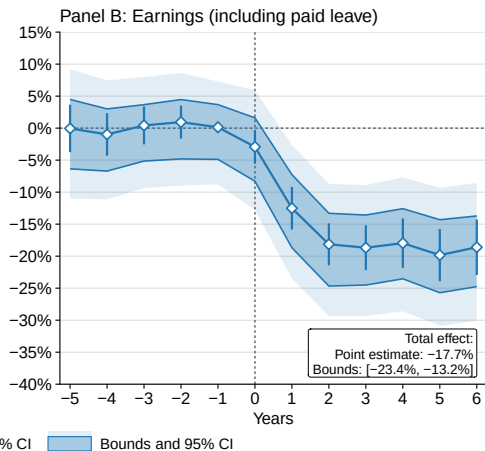
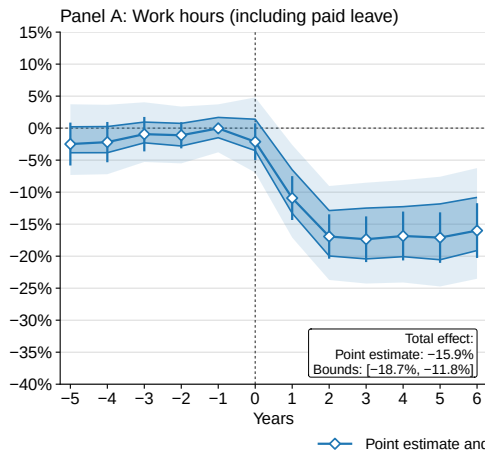
Parenthood



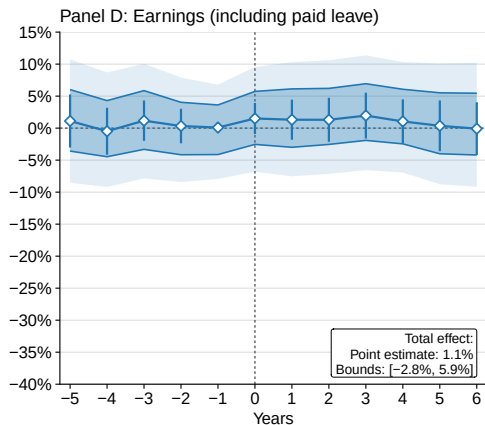
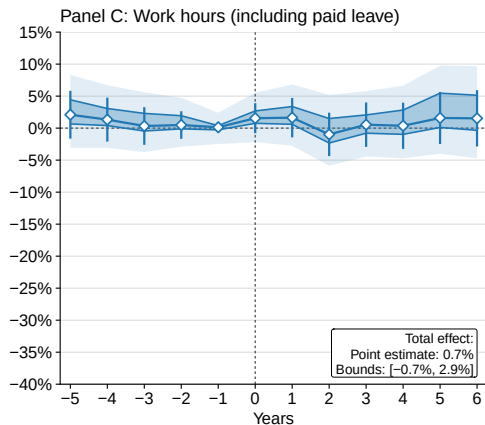
Number of Children



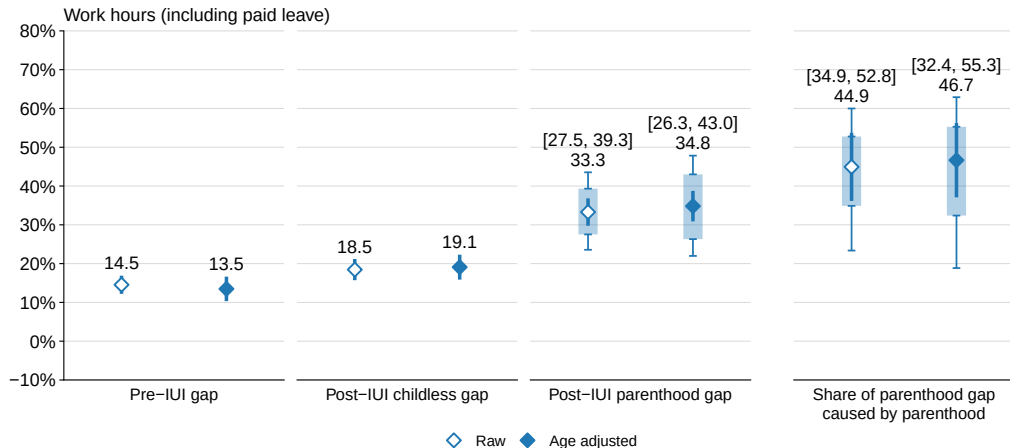
Effects of Parenthood: Women



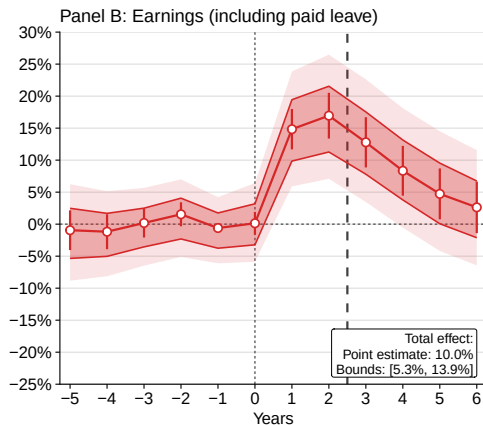
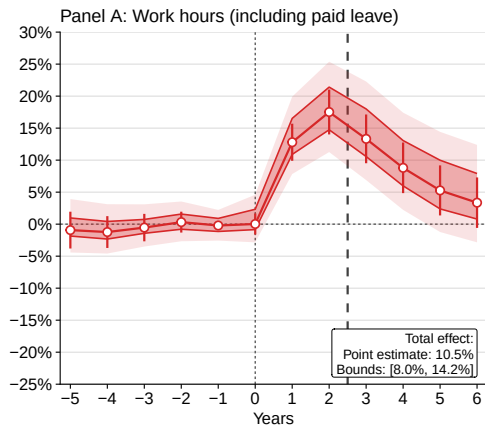
Effects of Parenthood: Men



Parenthood and the Within-Couple Gender Gap

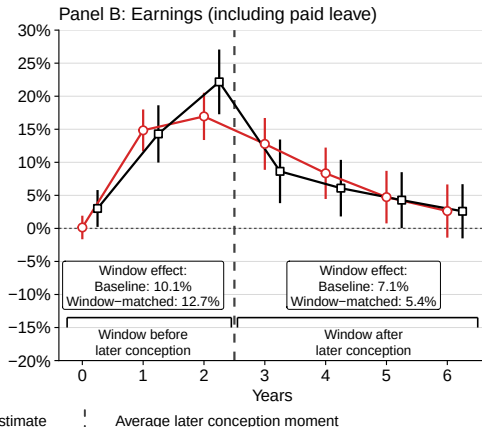
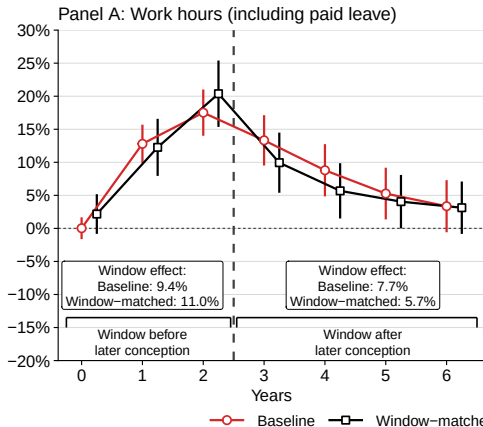


Effects of Later Parenthood: Women



—○— Point estimate and 95% CI ■ Bounds and 95% CI | Average delay

Effects of Later Parenthood: Matched Windows



Robustness, Mechanisms, and Heterogeneity

Selection on unobservables →

- Control for the recorded details of the treatment protocol
- Quantify how large selection can be and still reproduce balance on pre-IUI outcomes
- Use the non-rejected models to correct the estimates

Estimates change little.

Role of the intensive fertility margin →

- Effects of a second child, among couples undergoing IUI for a second child
- Effects of a multiple first birth, in the IUI and representative samples

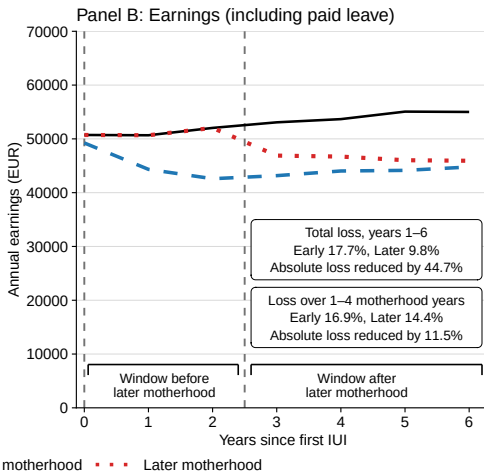
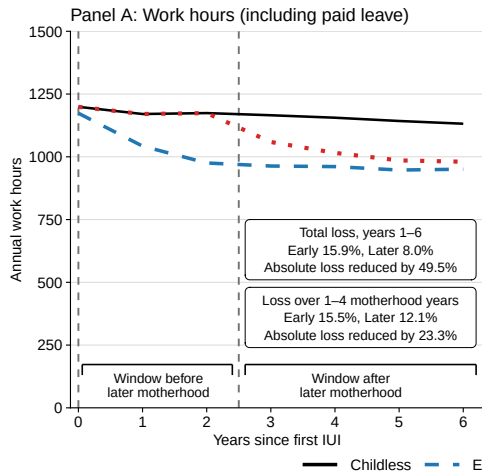
Little effect on women's work hours or earnings.

Heterogeneity →

- Larger benefits of delay for women who earned less before IUI, overall and relative to their partner, and for women younger at their first IUI

Differences are not explained by differences in timing or in the intensive margin.

Outcome Paths under the Three Scenarios



- Baseline: full-sample early-motherhood outcomes
- Assume effects are proportional for reliers and non-reliers
- Build career profiles for childlessness and for conception 2.5 years after the first IUI

Conclusion

Results:

- Motherhood accounts for up to half of gender inequality in hours and income
- Delayed childbearing mitigates women's losses

External relevance:

- IUI is common; the sample matches the population on observables
- Many women would undergo IUI if natural conception failed →

Comparison with existing work:

- Larger than Scandinavian IVF estimates; similar to contraceptive-failure estimates
- Not purely methodological: existing methods yield similar or larger effects →

Policy:

- Delaying may mitigate gender inequality, but reduce fertility and raise health risks
- Some gaps might persist even in the absence of children

How Existing Work Isolates Parenthood Effects

Parenthood effects may depend on current lifecycle stage and duration of parenthood:

$$\tau(\text{age, duration})$$

- **Age** = current career stage
- **Duration** = first child age
- **Age x Duration** = career stage at first birth

A) Standard IV approach (Lundborg et al., 2017, 2024):

$$\tau(\text{age, duration}) = \tau(\text{age})$$

B) Recursive IV approach (Bensnes et al., 2023; Gallen et al., 2023):

$$\tau(\text{age, duration}) = \tau(\text{duration}) \quad (\text{and homogeneous effects})$$

- A) requires that **duration** does not matter, B) that **career stage** does not matter
- Both A) and B) require that career stage at parenthood does not matter
- Estimates differ substantially across approaches (Bensnes et al., 2023)

Bounding τ_{ATR}

Construct the moment:

$$m^L(G, \eta^0) = Y 1_{\{Y < q(r(X_1), X_1)\}} \frac{Z_1}{e_1(X_1)} - Y(1 - D) \prod_{j=1}^A \frac{(1 - Z_j)}{(1 - e_j(X_j))}$$

- G is the observed data vector
- η^0 contains the following:
 - $e_j(X_j) = \Pr(Z_j = 1 | X_j)$
 - $q(r(X_1), X_1)$ is the $r(X_1)$ -th quantile of Y given X_1 and $Z_1 = 1$
 - $r(X_1)$ identifies the covariate-conditional relier share

Assumption (Conditional Sequential Unconfoundedness)

$(Y(k), R, W) \perp\!\!\!\perp Z_j \mid X_j$ for all j, k , and $X_j, A \geq j$.

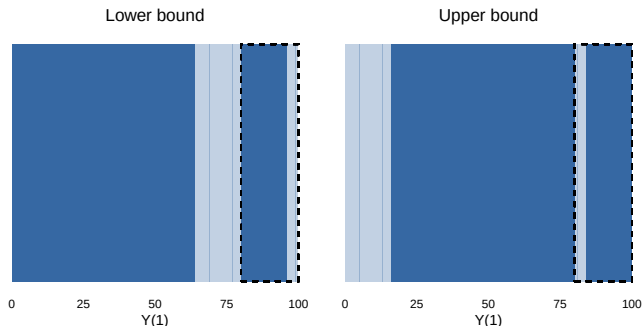
Theorem (Sharp Lower Bound)

Under conditional sequential unconfoundedness and regularity, the sharp lower bound on τ_{ATR} is $\mathbb{E}[m^L(G, \eta^0)] / \mathbb{E}[r(X_1)]$.

Intuition: Motherhood Outcome Y^E —Covariates

Pre-IUI covariates can help narrow the bounds:

- Can identify relier share at each covariate value
- Baseline bounds assume extreme scenarios where reliers have highest or lowest treated outcomes
- These distributions of treated outcomes might be inconsistent with conditional relier shares



Estimating the Bounds

Distribution of $m^L(G, \eta^0)$ is complicated by $q(r(X_1), X_1)$

- Semenova (2023) addresses a closely related inference challenge
- Double/debiased machine learning approach
 1. Adjust $m^L(G, \eta^0)$ to make it insensitive to small error in $q(r(X_1), X_1)$
 2. Sample splitting
- Asymptotic inference as if $q(r(X_1), X_1)$ was known

New moment:

$$\psi^L(G, \xi^0) = m^L(G, \eta^0) + \text{corr}(G, \xi^0)$$

Identifies same parameter:

$$\mathbb{E}[\psi^L(G, \xi^0)] = \mathbb{E}[m^L(G, \eta^0)]$$

Insensitive to estimation error in $q(r(X_1), X_1)$:

$$\partial_{q(\cdot)} \mathbb{E}[\psi^{L+}(G, \xi_r) | X_1] |_{\xi_r = \xi_r^0} = 0 \text{ a.s.}$$

Assisted Conception Procedures

- IUI (main procedure): sperm injected into uterus
 - Minimally invasive, primary ACP in most countries
 - “Free” in NL
- IVF (secondary procedure): embryo inserted into uterus
 - Invasive treatment, performed under sedation/anesthesia
 - Eggs retrieved through the vaginal wall using a specialized needle
 - In NL, first 3 free; each subsequent costs between 1000 and 4000 EUR

Institutions

- Dutch family friendly policies similar to OECD average
 - 16 weeks of fully paid pregnancy+maternity leave
 - 1 week of paternity leave
 - Average time in child care similar to OECD average
 - Net child care cost 10% median household income
- Dutch employment intensity similar to OECD average
 - Employment among parents and non-parents relatively high
 - Part time work much more common
 - Approximately 15% two-parent families have both partners working part-time

Data

Administrative data from Statistics Netherlands

- Comprehensive hospital records cover fertility treatments from 2012 to 2017: procedure date and type
 - Success imputed as having child born within 10 months
- Tax records cover work hours and income from 2011 to 2023
 - Include maternity leave and pay
 - Main bounds account for uncertainty around actual work hours
- Birth dates, legal family connections, cohabitation
- Dispensed medication registry

Main sample: cohabiting opposite-sex couples undergoing IUI for their first child between 2013 and 2016: 15,523

Table 1: First IUI Outcomes and Descriptives

	Success (1)	Fail (2)	Dif. (1)-(2)	IPW dif. (1)-(2) cond.	Rep. (5)	Suc. vs rep. (1)-(5)
Work (W)	0.912 [0.283]	0.916 [0.277]	-0.004 (0.008)	-0.009 (0.008)	0.936 [0.244]	-0.024 (0.007)
Work (P)	0.894 [0.307]	0.885 [0.319]	0.009 (0.009)	0.002 (0.009)	0.897 [0.304]	-0.002 (0.008)
Hours (W)	1300.012 [547.832]	1298.876 [558.316]	1.136 (15.730)	-1.951 (16.119)	1310.923 [544.468]	-10.911 (14.554)
Hours (P)	1513.337 [635.121]	1494.541 [656.050]	18.796 (18.457)	3.345 (19.041)	1497.603 [651.043]	15.734 (17.403)
Earn. 1000s EUR (W)	29.358 [18.000]	29.648 [18.911]	-0.290 (0.531)	0.203 (0.561)	26.555 [15.989]	2.803 (0.427)
Earn. 1000s EUR (P)	38.082 [25.425]	38.060 [26.525]	0.022 (0.745)	0.322 (0.774)	33.862 [24.148]	4.220 (0.646)
Bachelor deg. (W)	0.512 [0.500]	0.494 [0.500]	0.018 (0.014)		0.518 [0.500]	-0.007 (0.013)
Bachelor deg. (P)	0.425 [0.494]	0.410 [0.492]	0.014 (0.014)		0.430 [0.495]	-0.005 (0.013)
Age (W)	31.373 [3.889]	32.060 [4.265]	-0.687 (0.119)		28.840 [3.896]	2.533 (0.104)
Age (P)	34.088 [4.968]	34.856 [5.500]	-0.768 (0.154)		31.415 [4.803]	2.673 (0.128)
Observations	1,411	11,323			171,180	
Joint <i>p</i> -val.			0.001	0.955		0.000

Note: *Success* – average among women whose first IUI succeeded; *Fail* – average among women whose first IUI failed; *Dif.* – difference between *Success* and *Fail*; *IPW dif.* – difference adjusted for age and education using inverse probability weights from the baseline specification; *Rep.* – average in representative sample of women who conceived their first child without assisted conception procedures; *Suc. vs rep.* – difference between *Success* and *Rep.*. Reference year: year of first IUI (IUI sample); 9 months before first birth (representative sample). IUI sample: women who underwent intrauterine insemination for their first child between 2013 and 2016, with no prior assisted conception procedures, cohabiting with a male partner in the year prior to the reference year. Representative sample: women with no assisted conception procedures before first birth, cohabiting with a male partner in the year prior to the reference year, with reference year between 2013 and 2016. Labor market outcomes measured in the year before the reference year; age measured in the reference year. *Bachelor deg.* – indicator for completing a bachelor's degree. *Earn.* – earnings, (W) – woman, (P) – partner. Standard deviations in brackets. Standard errors in parentheses.

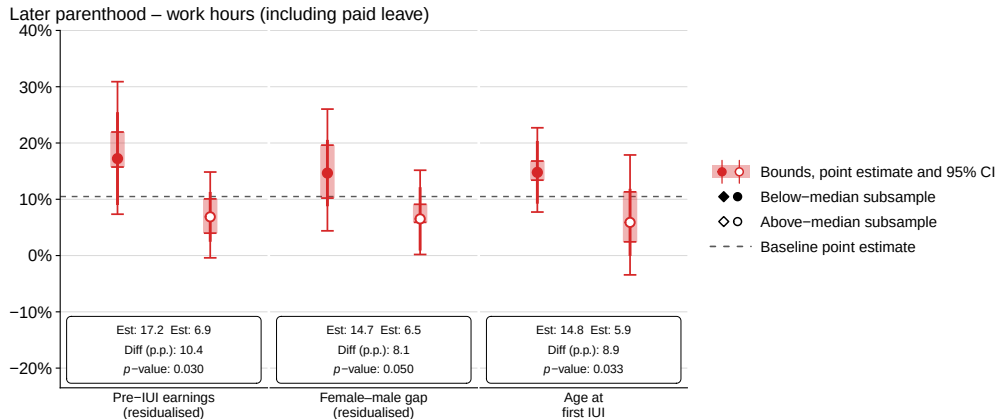
Balance in Subsequent ACPs

Table 2: Balance in Later ACPs

	Z ₂	Z ₃	Z ₄	Z ₅	Z ₆	Z ₇	Z ₈	Z ₉	Z ₁₀
Work (W)	0.009 (0.010)	-0.004 (0.011)	0.022 (0.011)	0.014 (0.012)	0.039 (0.012)	-0.003 (0.017)	-0.011 (0.018)	0.022 (0.019)	0.030 (0.024)
Work (P)	0.006 (0.010)	0.016 (0.010)	0.012 (0.012)	0.020 (0.012)	-0.004 (0.015)	-0.004 (0.015)	-0.019 (0.019)	0.017 (0.020)	0.030 (0.027)
Hours (W)	32.885 (18.721)	-4.482 (20.032)	52.999 (21.045)	41.332 (22.686)	81.957 (25.131)	11.894 (31.187)	-18.836 (32.937)	72.659 (38.210)	24.819 (48.490)
Hours (P)	21.655 (21.018)	24.730 (21.089)	23.756 (23.574)	38.965 (25.255)	9.666 (30.585)	-6.580 (31.513)	-28.458 (37.976)	30.525 (44.856)	43.722 (52.821)
Income 1000s € (W)	1.481 (0.615)	-0.015 (0.624)	1.685 (0.767)	1.802 (0.830)	2.086 (0.913)	0.150 (1.000)	-0.043 (1.092)	0.866 (1.234)	-0.444 (1.629)
Income 1000s € (P)	-0.749 (0.835)	1.002 (0.912)	2.040 (1.066)	0.800 (1.115)	0.774 (1.424)	0.025 (1.424)	0.259 (1.563)	-0.324 (1.737)	0.149 (2.203)
Observations	12,974	10,774	8,726	6,977	5,411	3,944	2,723	1,850	1,174
Joint <i>p</i> -val.	0.175	0.976	0.234	0.303	0.140	1.000	0.956	0.704	0.917

Note: Each column describes the difference in average characteristics between women for whom the respective ACP succeeds and those for whom it fails, among those who undergo the procedure, using inverse probability weights for each ACP following the main specification. Labor market outcomes and age measured year before first treatment. (W) - woman, (P) - partner. Standard errors in parentheses.

Heterogeneity in Timing Effects



The Role of the Intensive Fertility Margin

Parenthood is absorbing, so later births are part of both the early and later states.

- The intensive margin is therefore one channel through which timing could operate

Two exercises assess the role of the intensive margin:

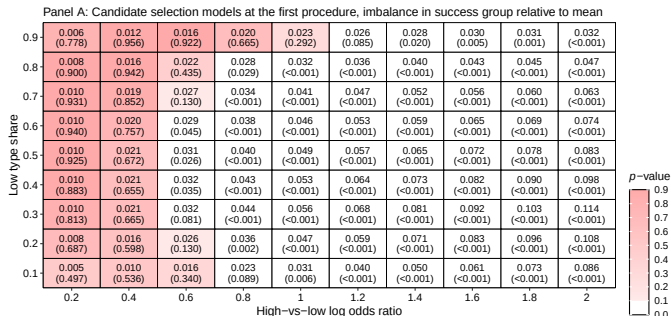
1. Effects of conception among parents undergoing IUI for a second child
 - Fertility increases by 0.76, yet there is little effect on women's work hours or earnings
2. Effects of a multiple first birth, in the IUI sample and in a representative sample
 - Fertility increases by 0.45, yet there is little effect on women's work hours or earnings
 - Main estimates are largely unchanged when multiple births are excluded

The intensive margin appears to play a limited role. [Back](#)

Sensitivity: Selection on Unobservables

1. Balance on pre-IUI outcomes is the strongest evidence Balance
 - Success at each attempt is uncorrelated with past outcomes conditional on age
2. I introduce a procedure that assumes latent heterogeneity in success likelihood Candidate models
 - Assign types from realized post-IUI motherhood outcomes so as to maximize selection
 - For each candidate model, test whether the imbalance it implies can still reproduce the observed pre-IUI balance
 - Models implying strong imbalance cannot
3. Use the non-rejected models to adjust the main estimates Adjusted estimates
 - Adjusted estimates stay close to the baseline; conclusions unchanged
4. Condition on the recorded details of the treatment protocol Protocol robustness
 - Re-estimate on any recorded protocol, on a summary of it, and on its full detail
 - Estimates change little

Candidate Selection Models



Panel B: Largest imbalance not rejected in success group relative to mean, by procedure

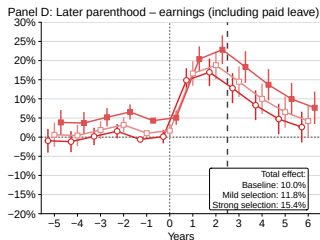
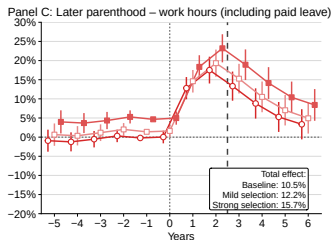
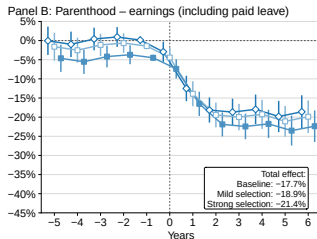
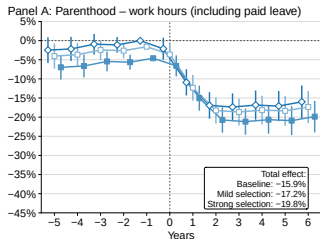
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	Pooled
0.027 (0.130)	0.018 (0.179)	0.017 (0.145)	0.016 (0.131)	0.030 (0.106)	0.035 (0.191)	0.028 (0.102)	0.032 (0.124)	0.021 (0.131)	0.054 (0.131)	0.009 (0.264)

Panel C: Largest imbalance not rejected in failure group relative to mean, by procedure

(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	Pooled
-0.003 (0.130)	-0.002 (0.179)	-0.002 (0.145)	-0.002 (0.131)	-0.004 (0.106)	-0.005 (0.191)	-0.006 (0.102)	-0.006 (0.124)	-0.005 (0.131)	-0.010 (0.131)	-0.001 (0.264)

Each cell is a model of latent heterogeneity: the implied imbalance in the success group, with the p -value of the test that it reproduces pre-IUI balance.

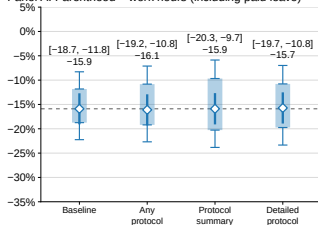
Estimates Adjusted for Non-Rejected Selection Models



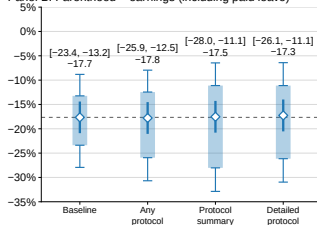
$\diamond$ Baseline $\square$ Mild selection $\blacksquare$ Strong selection | Average delay

Robustness to Controlling for the Treatment Protocol

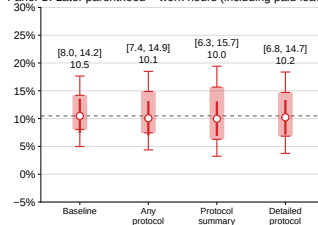
Panel A: Parenthood – work hours (including paid leave)



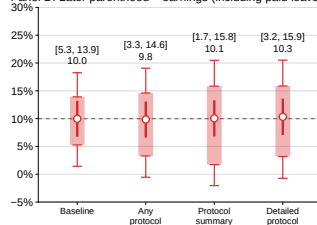
Panel B: Parenthood – earnings (including paid leave)



Panel C: Later parenthood – work hours (including paid leave)



Panel D: Later parenthood – earnings (including paid leave)



IUI Sample Representativeness: Observables

External relevance depends on whether the IUI sample resembles other parents

- Before parenthood, IUI parents worked slightly less and had somewhat lower income; differences are small [Table](#)
- IUI sample is 2.5 years older at first birth

Partly due to the one-year natural-attempt requirement before IUI

[Back](#)

IUI Sample Representativeness: Unobservables

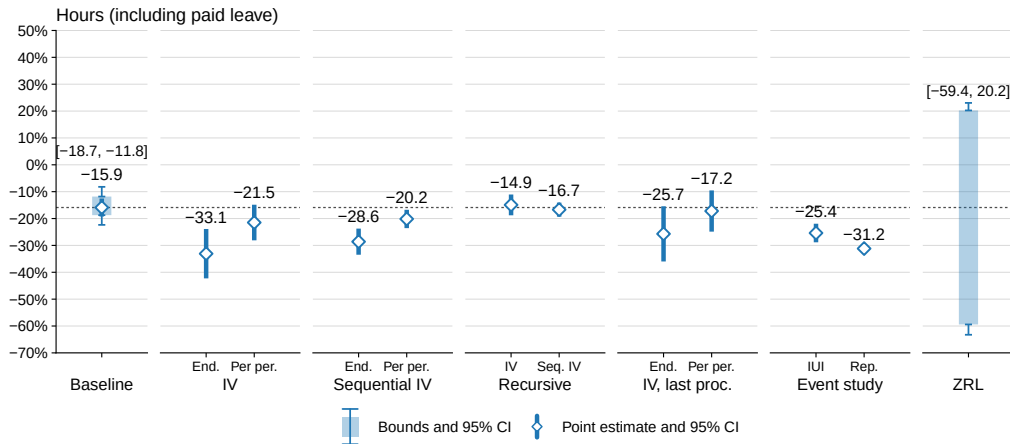
What about unobservables? E.g. stronger desire for family?

- 5% of Dutch mothers underwent IUI before conceiving; selection on unobservables may be substantial

IUI parents can be seen as draws from a population of potential IUI users

- IUI is initiated after one year of unsuccessful natural attempts
- Use medical fecundity rates to obtain a lower bound on how many women would have undergone IUI absent earlier natural conception
- Accounting for the age profile, for each IUI user, roughly four similar women conceive naturally before reaching IUI initiation
- Calculations suggest the sample is representative of at least $\sim 25\%$ of parents

Comparison with Existing Estimators



References I

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